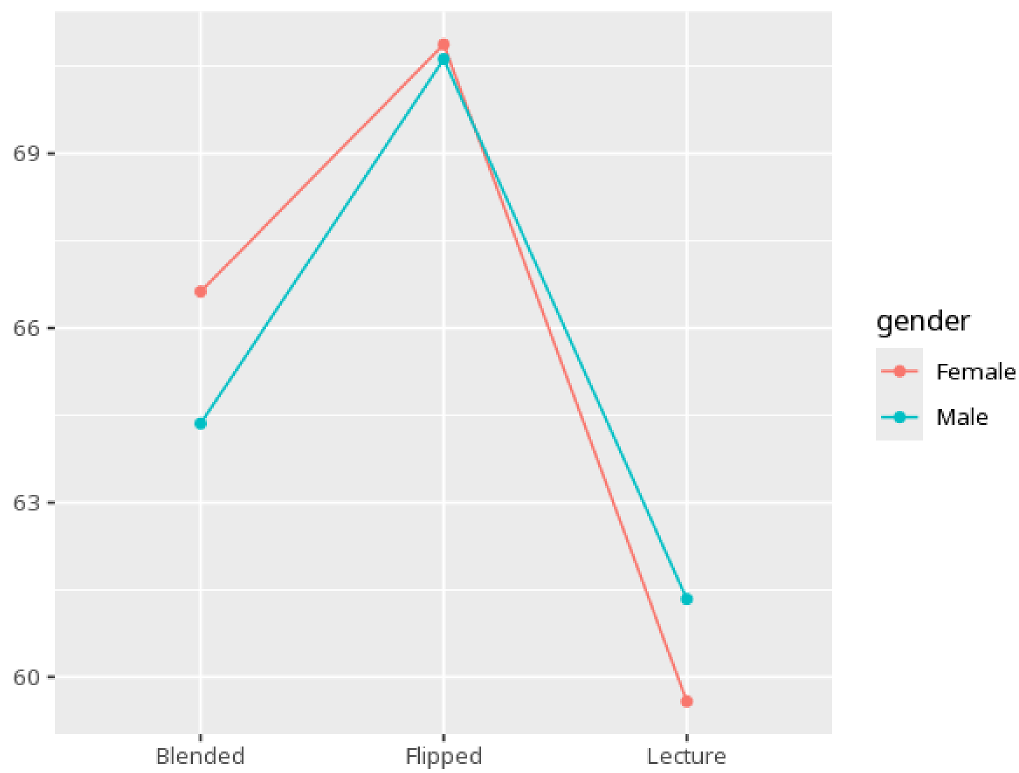
1 Factorial ANOVA

Factor A × B	N	Mean	Std. Dev.
Blended × Female	40	66.63	6.73
Flipped × Female	40	70.88	6.55
Lecture × Female	40	59.58	7.45
Blended × Male	40	64.36	6.86
Flipped × Male	40	70.62	6.69
Lecture × Male	40	61.34	7.67

Source	SS	df	MS	F	p	partial η^2 [95% CI]
teaching_method	4234.40	2	2117.20	43.14	<.001	0.27 [0.19, 1.00]
gender	3.85	1	3.85	0.08	.780	0.00 [0.00, 1.00]
teaching_method × gender	162.25	2	81.12	1.65	.194	0.01 [0.00, 1.00]

Contrast	Mdiff	SE	padj	95% CI
Blended - Flipped	−5.26	1.11	<.001	[−7.87, −2.64]
Blended - Lecture	5.03	1.11	<.001	[2.42, 7.64]
Flipped - Lecture	10.29	1.11	<.001	[7.68, 12.90]



Each factor's F/p is its main effect; the A×B row tests whether the effect of one factor depends on the other. A significant interaction usually takes priority - read it from the interaction plot before the main effects. A significant effect for any factor with 3+ levels tells you the levels differ somewhere, not which ones - use the post-hoc / simple-effects (emmeans) table to find the specific pairs. Your significance threshold (α) is 0.05.

A two-way ANOVA gave A×B interaction F(2,234)=1.65, p = .194, partial η^2=.01 [.00, 1.00].

Statistical basis: Factorial ANOVA (main effects + interaction) (Fisher, R. A. (1925). Statistical Methods for Research Workers. Oliver & Boyd.); Estimated marginal means for interaction / simple-effects follow-up (Lenth RV (2024). emmeans: Estimated Marginal Means, aka Least-Squares Means. R package.); η^2 effect-size benchmarks (Cohen, J. (1988). Statistical Power Analysis for the Behavioral Sciences (2nd ed.). Routledge.).